Hitbhai Lunagariya

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Profile

A third-year B.Tech student in Engineering Physics at SVNIT (CGPA 9.34/10, Rank 1) with a Minor in AI, building **NLP and data** systems across the full pipeline. Contributed to a **RAG-based** report-generation pipeline (Python, LangChain, Qdrant) as an AI/ML intern at EnviroWealth, and built **Ask the Data**, a deployed natural-language-to-**SQL** tool with an evaluation harness. Also trained and deployed computer vision models (**YOLOv8, OpenVINO, ROS2**) on a rover team that ranked **9th internationally** at IRC'26, the only team representing NIT.

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT) Aug 2024 – May 2028 (Expected)
B.Tech in Engineering Physics *CGPA: 9.34/10 (Rank 1)*
Minor in Artificial Intelligence *CGPA: 10/10*

Experience

EnviroWealth | AI/ML Intern | *Tesseract, LangChain, Streamlit, Qdrant* May 2026 – Jul 2026

- Contributed to a **RAG-based pipeline** for generating reports (OCR using **Tesseract**, chunking using Docling, vector database using **Qdrant**) for automated generation of BRSR sustainability reports
- Built a standalone chatbot prototype using **Streamlit** and the **Gemini API** to explore conversational report generation.

Projects

International Rover Challenge (IRC'26) | *Python, OpenCV, YOLOv8, ROS2* Oct 2025 – Feb 2026

- Developed a real-time computer vision pipeline for the IRC'26 Reconnaissance and Delivery Operation (RADO) mission, enabling fully autonomous cone detection, classification, and navigation on a rover.
- Expanded a Kaggle dataset from **4** to **6 cone classes** via **HSV-based synthetic generation**,trained a custom **YOLOv8** achieving reliable **mAP** across varied lighting conditions.
- Optimized YOLOv8 model via **OpenVINO** conversion for CPU-only deployment on Intel NUC, resolving real-time latency bottleneck and enabling reliable real-time cone tracking.
- Built an **ArUco marker detection** and pose estimation system for rover navigation, in the Autonomous Mission.
- Deployed the model inside a Robot Operating System (**ROS2**) to stream cone coordinates to the navigation stack.

Ask the Data | *Python, SQLite, sqlglot, Gemini API, Streamlit* Nov 2026

- Built a natural-language-to-**SQL** tool over UNHCR and World Bank data in a linked three-table **SQLite** database, with a **Streamlit** interface deployed on Streamlit Community Cloud.
- Designed **LLM prompts** (Gemini API) using the table schema, column descriptions and real category values. compared zero-shot, few-shot and error-retry strategies.
- Added two safety layers: a **sqlglot** validator (single SELECT, allowed tables, row limit) and a read-only SQLite connection with a query time limit.

Technical Skills

Core Areas: Natural Language Processing, RAG, Prompt Engineering, Computer Vision, Object Detection, Deep Learning, Machine Learning
Databases & Data: SQL (SQLite), Qdrant (vector database), sqlglot, pandas
Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, YOLOv8, LangChain, LangGraph, OpenVINO, Streamlit, Gemini API
Programming Languages: Python, SQL, C, C++
Tools & Robotics: Git/GitHub, pytest, GitHub Actions, ROS2, ESP32, Arduino

Leadership & Achievements

International Rover Challenge 2026 (IRC'26) — 9th Internationally
Branch Councilor – Engineering Physics, SVNIT (2024–2025)
Member – Drishti Technical Chapter (2025–Present)
Executive – Alumni Training & Placement Affairs Council SVNIT (2025–Present)